

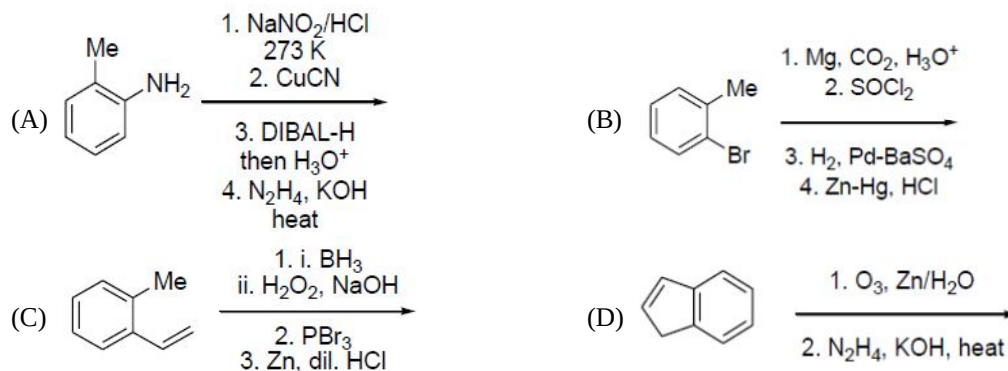
PART II: CHEMISTRY

SECTION 1

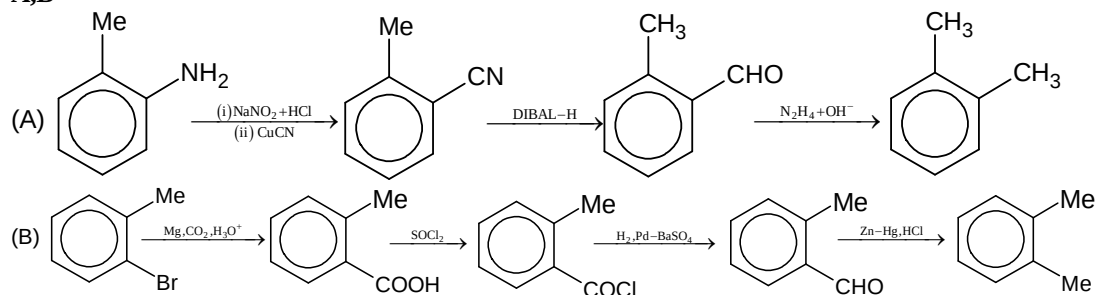
- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

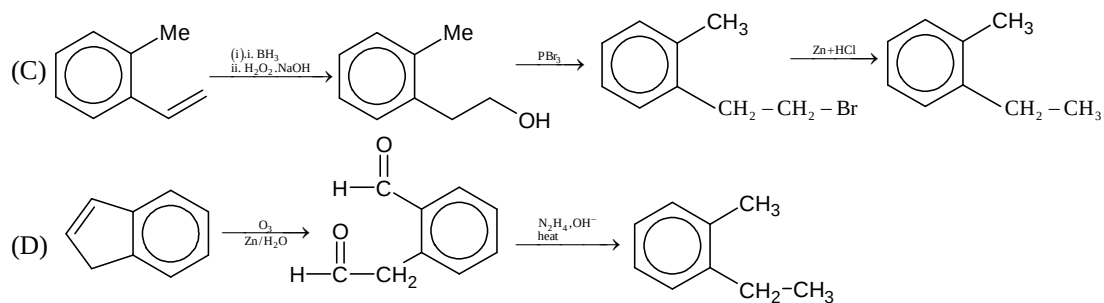
<i>Full Mark</i>	:	+4	If only (all) the correct option(s) is(are) chosen;
<i>Partial Marks</i>	:	+3	If all the four options are correct but ONLY three options are chosen;
<i>Partial Marks</i>	:	+2	If three or more options are correct but ONLY two options are chosen, both of which are correct;
<i>Partial Marks</i>	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
<i>Zero Marks</i>	:	0	If unanswered;
<i>Negative Marks</i>	:	-2	In all other cases.
- For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then
 - choosing ONLY (A), (B) and (D) will get +4 marks;
 - choosing ONLY (A) and (B) will get +2 marks;
 - choosing ONLY (A) and (D) will get +2 marks;
 - choosing ONLY (B) and (D) will get +2 marks;
 - choosing ONLY (A) will get +1 mark;
 - choosing ONLY (B) will get +1 mark;
 - choosing ONLY (D) will get +1 mark;
 - choosing no option(s) (i.e. the question is unanswered) will get 0 marks and choosing any other option(s) will get -2 marks.

1. The reaction sequence(s) that would lead to *o*-xylene as the major product is(are)

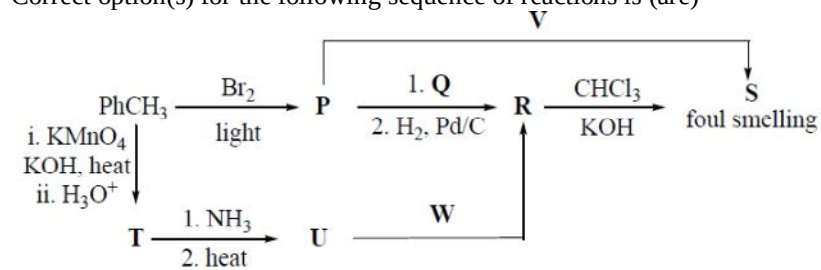


Sol.: A,B



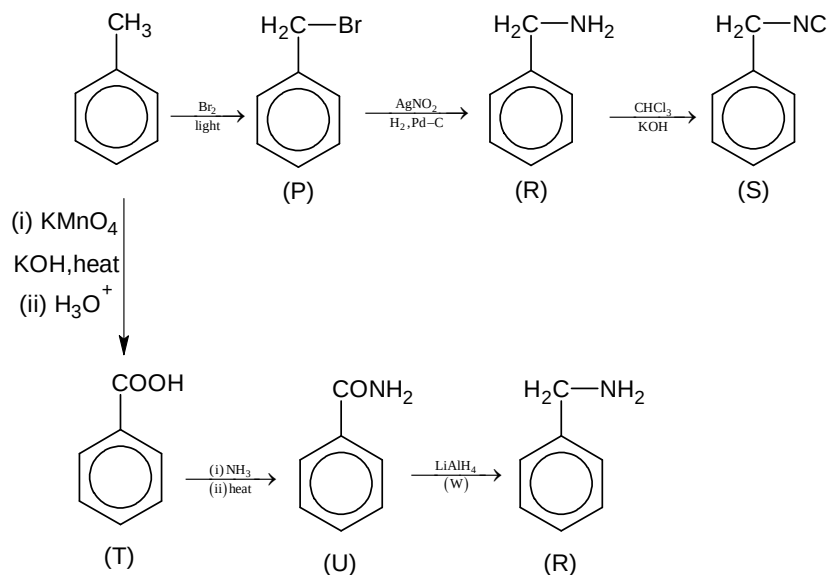


2. Correct option(s) for the following sequence of reactions is (are)



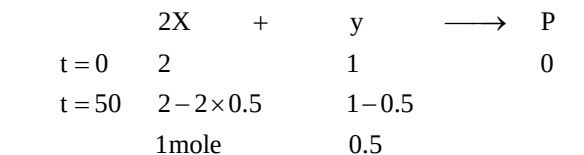
- (A) Q = KNO₂, W = LiAlH₄ (B) R = benzenamine, V = KCN
(C) Q = AgNO₂, R = phenylmethanamine (D) W = LiAlH₄, V = AgCN

Sol.: C, D



- *3. For the following reaction $2X + Y \xrightarrow{k} P$ the rate of reaction is $\frac{d[P]}{dt} = k[X]$. Two moles of **X** are mixed with one mole of **Y** to make 1.0 L of solution. At 50 s, 0.5 mole of **Y** is left in the reaction mixture. The correct statement(s) about the reaction is(are)
- (Use: $\ln 2 = 0.693$)
- (A) The rate constant, k , of the reaction is $13.86 \times 10^{-4} \text{ s}^{-1}$
- (B) Half-life of **X** is 50s
- (C) At 50 s, $-\frac{d[X]}{dt} = 13.86 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$.
- (D) At 100 s, $-\frac{d[Y]}{dt} = 3.46 \times 10^{-3} \text{ mol L}^{-1} \text{ s}^{-1}$.

Sol.: B,C,D



$$\text{rate} = -\frac{1}{2} \frac{dx}{dt} = -\frac{dy}{dt} = \frac{dP}{dt} = K[X]$$

$$-\frac{1}{2} \frac{dx}{dt} = K[X]$$

$$-\frac{dx}{dt} = 2K[X] = K^1[X]$$

Half life is $t = 50 \text{ sec}$

$$2K = \frac{0.693}{50}$$

$$K = \frac{0.6932}{100} = 6.332 \times 10^{-3}$$

t = 50 sec

$$-\frac{dx}{dt} = 2K[X]$$

$$-\frac{dx}{dt} = 2 \times 6.332 \times 10^{-3} \times 1 = 13.864 \times 10^{-3} \text{ mole / L / Sec}$$

$$-\frac{dy}{dt} = K[X] = 6.332 \times 10^{-3} \left(\frac{1}{2} \right) = 3.46 \times 10^{-3} \text{ mole / L / Sec}^{-1}$$

4. Some standard electrode potentials at 298 K are given below:

$$\text{Pb}^{2+}/\text{Pb} - 0.13 \text{ V}$$

$$\text{Ni}^{2+}/\text{Ni} - 0.24 \text{ V}$$

$$\text{Cd}^{2+}/\text{Cd} - 0.40 \text{ V}$$

$$\text{Fe}^{2+}/\text{Fe} - 0.44 \text{ V}$$

To a solution containing 0.001 M of X^{2+} and 0.1 M of Y^{2+} , the metal rods **X** and **Y** are inserted (at 298 K) and connected by a conducting wire. This resulted in dissolution of **X**. The correct combination(s) of **X** and **Y**, respectively, is(are)

(Given: Gas constant, $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$,

Faraday constant, $F = 96500 \text{ C mol}^{-1}$)

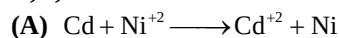
(A) Cd and Ni

(B) Cd and Fe

(C) Ni and Pb

(D) Ni and Fe

Sol.: A,B,C



$$E_{\text{cell}} = 0.40 + (-24) - \frac{0.0591}{2} \log \frac{0.001}{0.1}$$

$$= 0.16 + \frac{0.0591}{2} \times 2 = 0.16 + 0.0591 = 0.2191 (+ve)$$

(B) $E_{\text{cell}} = 0.40 + (-0.44) - \frac{0.591}{2} \log \frac{0.01}{0.1}$

$$= -0.04 + \frac{0.591}{2} \times 2 = -0.04 + 0.591 = 0.551 (+ve)$$

(C) $E_{\text{cell}} = 0.24 + (-0.13) + \frac{.0591}{2} \times 2$

$$= 0.11 + 0.06 = 0.17 (+ve)$$

(D) $E_{\text{cell}} = 0.24 + (-0.44) + \frac{0.0591}{2} \times 2$

$$= -0.20 + 0.06 = -0.14 (-ve)$$

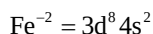
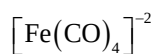
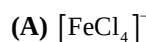
5. The pair(s) of complexes wherein both exhibit tetrahedral geometry is(are)

(Note: py = pyridine)

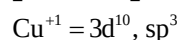
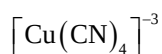
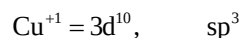
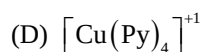
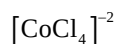
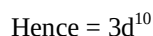
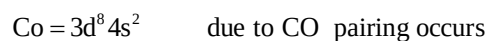
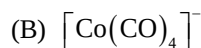
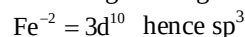
Given: Atomic numbers of Fe, Co, Ni and Cu are 26, 27, 28 and 29, respectively)

- (A) $[\text{FeCl}_4]^-$ and $[\text{Fe}(\text{CO})_4]^{2-}$ (B) $[\text{Co}(\text{CO})_4]^-$ and $[\text{CoCl}_4]^{2-}$
- (C) $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}(\text{CN})_4]^{2-}$ (D) $[\text{Cu}(\text{py})_4]^+$ and $[\text{Cu}(\text{CN})_4]^{3-}$

Sol.: A, B, D

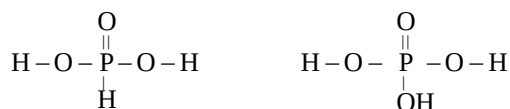
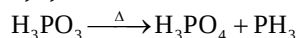


CO strong field ligand pairing occurs



6. The correct statement(s) related to oxoacids of phosphorous is(are)
- (A) Upon heating, H_3PO_3 undergoes disproportionation reaction to produce H_3PO_4 and PH_3 .
- (B) While H_3PO_3 can act as reducing agent, H_3PO_4 cannot.
- (C) H_3PO_3 is a monobasic acid.
- (D) The H atom of P–H bond in H_3PO_3 is not ionizable in water.

Sol.: A,B,D



P–H bond is responsible for its reducing character, H_3PO_4 does not have.

H_3PO_3 is a dibasic acid in oxyacids of phosphorus O–H bond is ionisable whereas P–H bond is non ionizable.

SECTION 2

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place using the mouse and the on-screen virtual numeric keypad.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

<i>Full Mark</i>	:	+2	If ONLY the correct numerical value is entered at the designated place;
<i>Zero Marks</i>	:	0	In all other cases.

Question Stem for Question Nos. 7 and 8

Question Stem

At 298 K, the limiting molar conductivity of a weak monobasic acid is $4 \times 10^2 \text{ S cm}^2 \text{ mol}^{-1}$. At 298 K, for an aqueous solution of the acid the degree of dissociation is α and the molar conductivity is $y \times 10^2 \text{ S cm}^2 \text{ mol}^{-1}$. At 298 K, upon 20 times dilution with water, the molar conductivity of the solution becomes $3y \times 10^2 \text{ S cm}^2 \text{ mol}^{-1}$.

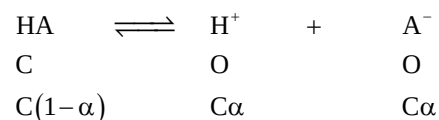
7. The value of α is _____.

Sol.: 0.22

$$\alpha_1 = \frac{\Lambda_m^c}{\Lambda_m^0} = \frac{y \times 10^2}{4 \times 10^2} = \frac{y}{4} = \alpha$$

On dilution conductivity increases three times

$$\alpha_2 = \frac{3y \times 10^2}{4 \times 10^2} = 3\alpha_1 = 3\alpha$$



$$K_a = \frac{C\alpha^2}{1-\alpha}$$

Since temperature is constant K_a will be constant

$$\frac{C_1\alpha_1^2}{1-\alpha_1} = \frac{C_2\alpha_2^2}{1-\alpha_2}$$

$$\frac{C \times \alpha^2}{1-\alpha} = \frac{\left(\frac{C}{20}\right)(3\alpha)^2}{1-3\alpha}$$

$$\frac{1}{1-\alpha} = \frac{9}{20} - \frac{1}{(1-3\alpha)}$$

$$20 - 60\alpha = 9 - 9\alpha;$$

$$\alpha = \frac{11}{51} = 0.2156$$

$$\alpha = 0.22$$

8. The value of **y** is _____.

Sol.: 0.88

$$\alpha = \frac{y}{4}; \quad y = 4\alpha = 4 \times 0.22 = 0.88$$

Question Stem for Question Nos. 9 and 10

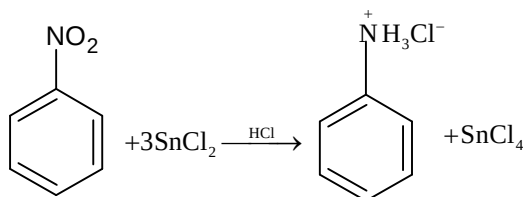
Question Stem

Reaction of **x** g of Sn with HCl quantitatively produced a salt. Entire amount of the salt reacted with **y** g of nitrobenzene in the presence of required amount of HCl to produce 1.29 g of an organic salt (quantitatively).

(Use Molar masses (in g mol⁻¹) of H, C, N, O, Cl and Sn as 1, 12, 14, 16, 35 and 119, respectively).

9. The value of **x** is _____.

Sol.: 3.57



$$\text{Moles of Sn} = \frac{x}{119}$$

$$\text{Moles of nitrobenzene} = \frac{x}{119} \times \frac{1}{3}$$

$$\text{Moles of anilium chloride} = \frac{x}{119} \times \frac{1}{3}$$

$$\text{Moles of nitrobenzene} = \frac{x}{357}$$

$$\frac{y}{123} = \frac{x}{357} \quad \dots (1)$$

$$\text{Moles of anilium chloride} = \frac{x}{357} \quad \dots (2)$$

$$\frac{1.29}{129} = \frac{x}{357}$$

$$x = 3.57$$

10. The value of **y** is _____.

Sol.: 1.23

$$y = 1.225 \cong 1.23$$

Question Stem for Question Nos. 11 and 12

Question Stem

A sample (5.6 g) containing iron is completely dissolved in cold dilute HCl to prepare a 250 mL of solution. Titration of 25.0 mL of this solution requires 12.5 mL of 0.03 M KMnO_4 solution to reach the end point. Number of moles of Fe^{2+} present in 250 mL solution is $x \times 10^{-2}$ (consider complete dissolution of FeCl_2). The amount of iron present in the sample is **y**% by weight.

(Assume: KMnO_4 reacts only with Fe^{2+} in the solution)

Use: Molar mass of iron as 56 g mol^{-1})

*11. The value of **x** is _____.

Sol.: 1.875

$$\text{meq of Fe}^{+2} = \text{meq of KMnO}_4$$

$$x \times 10^{-2} \times 1000 \times 1 = 12.5 \times 0.03 \times 5 \times 10$$

$$x = 1.875 \text{ mole}$$

*12. The value of **y** is _____.

Sol.: 18.75

$$\text{Moles of Fe}^{+2} = x \times 10^{-2} = 1.875 \times 10^{-2}$$

$$\text{wt. of Fe}^{+2} = 1.875 \times 10^{-2} \times 56$$

Hence percentage of Fe^{+2}

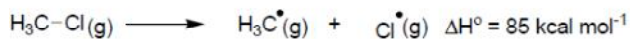
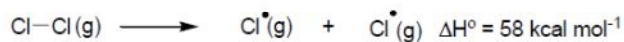
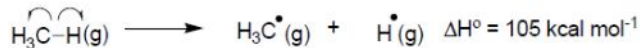
$$= \frac{1.875 \times 10^{-2} \times 56}{5.6} \times 100\% = 18.75\%$$

SECTION 3

- This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Mark : +3 If ONLY the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

Paragraph

The amount of energy required to break a bond is same as the amount of energy released when the same bond is formed. In gaseous state, the energy required for *homolytic cleavage* of a bond is called Bond Dissociation Energy (BDE) or Bond Strength. BDE is affected by s-character of the bond and the stability of the radicals formed. Shorter bonds are typically stronger bonds. BDEs for some bonds are given below:



- *13. Correct match of the **C-H** bonds (shown in bold) in Column J with their BDE in Column K is

Column J Molecule	Column K BDE (kcal mol ⁻¹)
(P) H - CH(CH ₃) ₂	(i) 132
(Q) H - CH ₂ Ph	(ii) 110
(R) H - CH = CH ₂	(iii) 95
(S) H - C ≡ CH	(iv) 88

(A) P - iii, Q - iv, R - ii, S - i

(B) P - i, Q - ii, R - iii, S - iv

(C) P - iii, Q - ii, R - i, S - iv

(D) P - ii, Q - i, R - iv, S - iii

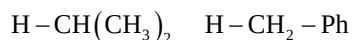
Sol.: A

As s character increases bond dissociation energy increases



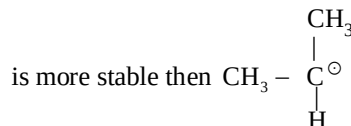
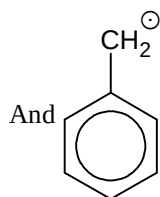
sp

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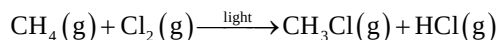


sp³

sp³



- *14. For the following reaction

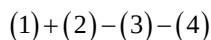
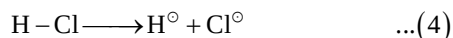
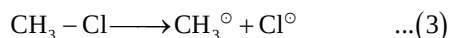
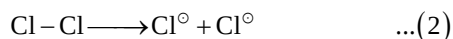
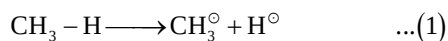


The correct statement is

- (A) Initiation step is exothermic with $\Delta H^\circ = -58 \text{ kcal mol}^{-1}$.
 (B) Propagation step involving $\cdot\text{CH}_3$ formation is exothermic with $\Delta H^\circ = -2 \text{ kcal mol}^{-1}$.
 (C) Propagation step involving CH_3Cl formation is endothermic with $\Delta H^\circ = +27 \text{ kcal mol}^{-1}$.
 (D) The reaction is exothermic with $\Delta H^\circ = -25 \text{ kcal mol}^{-1}$.

Sol.: D

$\text{CH}_4 + \text{Cl}_2 \xrightarrow{\text{light}} \text{CH}_3\text{Cl} + \text{HCl}$ this reaction is obtained from given reaction.

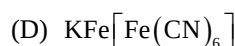
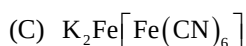
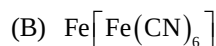
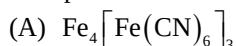


Hence $\Delta H = 105 + 58 - 85 - 103 = -25 \text{ K Cal / mole}$

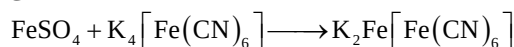
Paragraph

The reaction of $\text{K}_3[\text{Fe}(\text{CN})_6]$ with freshly prepared FeSO_4 solution produces a dark blue precipitate called Turnbull's blue. Reaction of $\text{K}_4[\text{Fe}(\text{CN})_6]$ with the FeSO_4 solution in complete absence of air produces a white precipitate **X**, which turns blue in air. Mixing the FeSO_4 solution with NaNO_3 , followed by a slow addition of concentrated H_2SO_4 through the side of the test tube produces a brown ring.

15. Precipitate **X** is

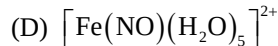
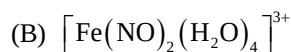
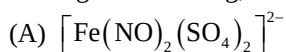


Sol.: C

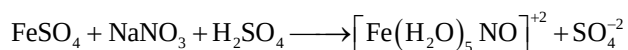


While ppt.

16. Among the following, the brown ring is due to the formation of



Sol.: D



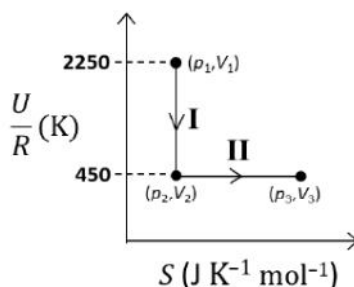
SECTION 4

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designed to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Mark	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

*17. One mole of an ideal gas at 900 K, undergoes two reversible processes, **I** followed by **II**, as shown below.

If the work done by the gas in the two process are same, the value of $\ln \frac{V_3}{V_2}$ —



(U : internal energy, S : entropy, p : pressure, V : volume, R : gas constant)

(Given: molar heat capacity at constant volume, $C_{v,m}$ of the gas is $\frac{5}{2}R$)

Sol.: 10

1st process is adiabatic since entropy is constant.

$$W_1 = \Delta U$$

$$\Delta U = 450R - 2250R = -1800R$$

$$W_1 = -1800R \quad \dots (1)$$

In 2nd process internal energy is constant it means it is a isothermal process.

$$W_2 = -2.303nRT \log \frac{V_3}{V_2} \quad \dots (2)$$

$$= -nRT \ln \frac{V_3}{V_2} \quad \dots (3)$$

Given, $n = 1$ mole, here temperature is unknown

$$U = nC_v T \quad \text{for process II}$$

$$450R = 1 \times \frac{5}{2} RT$$

$$T = \frac{450 \times 2}{5} = 180K$$

Equation (1) = equation (2)

$$W_1 = W_2$$

$$-1800R = -1 \times R \times 180 \ln \frac{V_3}{V_2}$$

$$\ln \frac{V_3}{V_2} = \frac{1800}{180} = 10$$

- *18. Consider a helium (He) atom that absorbs a photon of wavelength 330 nm. The change in the velocity (in cm s^{-1}) of He atom after the photon absorption is ____.
(Assume: Momentum is conserved when photon is absorbed.)
Use: Planck constant = 6.6×10^{-34} J s, Avogadro number = $6 \times 10^{23} \text{ mol}^{-1}$,
Molar mass of He = 4 g mol^{-1})

Sol.: 30

$$\lambda = \frac{h}{m(\Delta V)}$$

$$330 \times 10^{-9} = \frac{6.6 \times 10^{-34}}{\left(\frac{4 \times 10^{-3}}{6 \times 10^{23}} \right) \times \Delta V}$$

$$\Delta V = \frac{6.6 \times 6 \times 10^{23} \times 10^{-34}}{4 \times 10^{-3} \times 330 \times 10^{-9}}$$

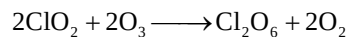
$$= 0.30 \text{ m/s}$$

$$= 0.30 \times 100 \text{ cm/s}$$

$$= 30 \text{ cm/s}$$

19. Ozonolysis of ClO_2 produces an oxide of chlorine. The average oxidation state of chlorine in this oxide is ____.

Sol.: 6



Average oxidation state of Cl atom is +6